# EcoExplain-VLM · experimental workflow

Kanha Tiger Reserve · dry season T0 Jan–Feb 2022 / T1 Jan–Feb 2023

**01 Multi-sensor inputs**

**Sentinel-2 optical**
RGB + 10 m bands
NDVI · EVI · NDMI · NBR

**Sentinel-1 SAR**
VV / VH backscatter
descending pass

**CHIRPS rainfall**
Window totals
T0 and T1

**ERA5-Land**
Temperature & soil moisture
anomalies vs 2015–2021

**NASADEM terrain**
Elevation · slope
static context

**02 Preprocessing**

**Earth Engine · matched composites, cloud mask, 640 m patches**

Study rectangle 80.433–81.050°E, 22.117–22.450°N · 1,200 candidate centres · 15 GeoTIFFs per patch · T0/T1 RGB chips for the VLM

**03 Labels and evidence**

**Paired RGB chips**
True-colour T0 and T1
visual input to every VLM

**Automatic class (gold)**
Percentiles of ΔNDVI on the pool
decline · stable · increase

**Structured evidence card**
Indices, SAR, climate, terrain
full card still has candidate_class=

**04 Dataset freeze**

**366 patches → train 256 / development 55 / test 55 (seed 42)**

Gold = candidate_class · majority baseline on test = 34.5% · same 55 test patches in every protocol below

**05 Three evaluation tracks (do not mix the scores)**

**A Zero-shot classification**
Condition C: T0/T1 RGB + card
Qwen2-VL-2B · PaliGemma-3B
LLaVA-Interleave-0.5B · BLIP-2
Full card and evidence_only

**B Class LoRA (smoke test)**
Qwen2-VL-2B · r = 8 · 1 epoch
16 training patches
100% run with leak is invalid

**C Explanations (label stripped)**
Free-form vs structured cite-only
Claims scored against the card
Optional 2B-on-2B explain LoRA